Alireza Karbalayghareh

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Research Interests

My research focuses on developing universal AI models of gene regulation, capturing cis-regulation, trans-regulation, and intercellular signaling by integrating multimodal, multiomic observational, spatial, time-resolved, and perturbational data from immune cells across diverse biological contexts to understand and engineer the immune system to prevent and treat disease.

Research Positions

Biohub, New York Jun. 2025–present

Senior Research Scientist, AI/ML

Memorial Sloan Kettering Cancer Center Sept. 2019–Jun. 2025

Postdoctoral Research Associate, Computational & Systems Biology Program
PI: Dr. Christina S. Leslie

Education

Texas A&M University Sept. 2014–Aug. 2019

Ph.D. in Electrical Engineering (Computational Biology Group)
Advisor: Dr. Edward R. Dougherty
Dissertation: *Optimal Bayesian Transfer Learning for Classification and Regression*

Sharif University of Technology Sept. 2011–Aug. 2014

M.Sc. in Electrical Engineering

Iran University of Science and Technology Sept. 2007–Aug. 2011

B.Sc. in Electrical Engineering, with honors

Publications

[*] Equal contribution.

[1] **A. Karbalayghareh**, B. Pelzer, C. R. Chin, A. M. Melnick, D. Barisic, and C. S. Leslie. Deep dynamical models of single-cell multiomic velocities predict loss-of-function and rescue perturbations in B cells. *Nature Biotechnology* (2026).

[2] A. R. Gschwind[*], K. S. Mualim[*], **A. Karbalayghareh**[*], ..., J. M. Engreitz. An encyclopedia of human enhancer–gene regulatory interactions. *Nature* **657**, 179–190 (2026).

[3] R. Yang, **A. Karbalayghareh**, and C. S. Leslie. HiC2Self: Self-supervised denoising for bulk and single-cell Hi-C contact maps. *Science Advances* **12**(30) (2026).

[4] B. Pelzer, C. Meydan, . . . , **A. Karbalayghareh**, . . . , A. M. Melnick. SPEN Loss Drives Extrafollicular Diffuse Large B-cell Lymphoma with Female-Specific Lethality and Therapeutic Vulnerabilities. *Cancer Discovery* **16**(8), 1666–1690 (2026).

[5] S. Tiwari, **A. Karbalayghareh**, and C. S. Leslie. Predicting the regulatory genome. *Nature Reviews Genetics* **26**, 659–660 (2025).

[6] V. R. Gao, R. Yang, A. Das, . . . , **A. Karbalayghareh**, . . . , C. S. Leslie. ChromaFold predicts the 3D contact map from single-cell chromatin accessibility. *Nature Communications* **15**, 9432 (2024).

[7] R. Yang*, A. Das*, V. R. Gao, **A. Karbalayghareh**, W. S. Noble, J. A. Bilmes, and C. S. Leslie. Epiphany: predicting Hi-C contact maps from 1D epigenomic signals. *Genome Biology* **24**, 134 (2023).

[8] **A. Karbalayghareh**, M. Sahin, and C. S. Leslie. Chromatin interaction–aware gene regulatory modeling with graph attention networks. *Genome Research* **32**(5), 930–944 (2022).

[9] **A. Karbalayghareh**, X. Qian, and E. R. Dougherty. Optimal Bayesian transfer learning for count data. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* (2021).

[10] E. Hajiramezanali, S. Zamani Dadaneh, **A. Karbalayghareh**, M. Zhou, and X. Qian. Bayesian multi-domain learning for cancer subtype discovery from next-generation sequencing count data. *Advances in Neural Information Processing Systems* **31** (NeurIPS, 2018).

[11] **A. Karbalayghareh**, X. Qian, and E. R. Dougherty. Optimal Bayesian transfer regression. *IEEE Signal Processing Letters* **25**(11), 1655–1659 (2018).

[12] **A. Karbalayghareh**, X. Qian, and E. R. Dougherty. Optimal Bayesian transfer learning. *IEEE Transactions on Signal Processing* **66**(14) (2018).

[13] **A. Karbalayghareh**, U. Braga-Neto, and E. R. Dougherty. Intrinsically Bayesian robust classifier for single-cell gene expression trajectories in gene regulatory networks. *BMC Systems Biology* **12**, 23 (2018).

[14] **A. Karbalayghareh**, U. Braga-Neto, and E. R. Dougherty. Classification of single-cell gene expression trajectories from incomplete and noisy data. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* (2017).

[15] **A. Karbalayghareh**, U. Braga-Neto, and E. R. Dougherty. Classification of Gaussian trajectories with missing data in Boolean gene regulatory networks. *IEEE International Conference on Acoustics, Speech and Signal Processing* (ICASSP), New Orleans, LA (2017).

[16] **A. Karbalayghareh**, U. Braga-Neto, J. Hua, and E. R. Dougherty. Classification of state trajectories in gene regulatory networks. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* (2016).

[17] **A. Karbalayghareh** and T. Hu. Inference of sparse gene regulatory network from RNA-seq time series data. *IEEE Global Conference on Signal and Information Processing* (GlobalSIP), Orlando, FL (2015).

Preprints

[P1] **A. Karbalayghareh**, E. Paull, and A. Califano. DoFormer: Causal Transformer for Gene Perturbation. *bioRxiv* (2026).

[P2] V. Rapolu, **A. Karbalayghareh**, B. Lee, W. Wong, and C. S. Leslie. scDIVA: semi-supervised integration and fine-grained annotation of tumor-immune single-cell atlases. *bioRxiv* (2026).

Selected Research Contributions

DoFormer: intervention-aware gene perturbation modeling Preprint P1

- Developed a Transformer for single-cell genomics that encodes gene interventions within attention, distinguishing observational and interventional regimes without requiring a directed acyclic graph.
- Improved prediction of held-out gene perturbations relative to baseline methods and transcriptomic foundation models using biologically informed training objectives.

DynaVelo: dynamical modeling of single-cell multiomics Publication 1

- Developed a generative neural ODE that jointly learns gene-expression and TF-motif-accessibility dynamics, including latent time and multiomic velocities.
- Modeled cyclic and branching germinal center B-cell dynamics and predicted loss-of-function and rescue perturbations, with experimental testing in B cells.

GraphReg: gene regulation through 3D genomic interactions Publication 8

- Developed graph attention models that integrate DNA sequence or epigenomic features with 3D chromatin interactions to predict gene expression.
- Used feature attribution to identify functional enhancers and distal TF motifs, and evaluated regulatory predictions against CRISPRi perturbation data.

ENCODE-rE2G: enhancer–gene regulatory interactions Publication 2

- Co-developed supervised models and software that integrate CRISPRi enhancer perturbations, epigenomic features, genomic distance, and 3D contacts to predict enhancer–gene interactions.
- Contributed to analyses of regulatory features and the ENCODE resource for linking noncoding variants to target genes and cell types.

scDIVA: annotation of tumor-immune single-cell atlases Preprint P2

- Co-developed semi-supervised approaches for immune-cell annotation across heterogeneous scRNA-seq datasets using domain-invariant variational autoencoders.

Bayesian transfer learning and regulatory dynamics Publications 9–17

- Developed Bayesian transfer classification and regression, multi-domain learning for count data, and classification and inference methods for gene-expression trajectories under incomplete observations and model uncertainty.

Research Leadership and Collaboration

- Lead a cross-functional project to develop universal AI models of gene regulation in immune cells by integrating large-scale, multimodal data generated across Biohub.
- Led the 4D Nucleome hackathon project, “Evaluating the effects of Hi-C and TFs on cis-regulation of gene expression” (Mar. 2024).
- Contributed to collaborative research in the NIH ENCODE, Impact of Genomic Variation on Function (IGVF), and 4D Nucleome (4DN) consortia (2019–2025).

Research Mentoring

Research mentorship of doctoral and master's students in computational biology. Tri-I denotes the Tri-Institutional Program in Computational Biology and Medicine.

Trainee	Program / training level	Dates
Sarthak Tiwari	Tri-I, Ph.D. student	2024–present
Viraj Rapolu	Tri-I, Ph.D. student	2021–2025
Rebecca Murray	Tri-I, Ph.D. student	Winter–spring 2025
Taykhoom Dalal	Tri-I, Ph.D. student	Winter–spring 2024
Jiaxin Li	Tri-I, Ph.D. student	Summer–fall 2022
Divya Koyyalagunta	Tri-I, Ph.D. student	Summer 2021
Changlin Wan	Purdue University, Ph.D. summer intern	Summer 2021
Brennan Lee	Tri-I, M.Sc. student	2020–2021

- Mentored Ph.D. students in developing Epiphany and ChromaFold, models for predicting 3D chromatin contacts from epigenomic and single-cell chromatin-accessibility data (Publications 6–7).

Teaching Experience

University of North Carolina at Chapel Hill Nov. 2022

Guest lecturer in computational biology, invited by Dr. Michael Love; virtual.

Honors and Awards

- Best Poster Award, Memorial Sloan Kettering Postdoctoral Research Symposium (Sept. 2023).
- Best Poster Award, IGVF Consortium Meeting (Sept. 2023).
- NSF Travel Grant, ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (Aug. 2017).
- Travel Grant, MidSouth Computational Biology and Bioinformatics Society Conference (Mar. 2017).
- Ranked 3rd among 120 graduating Electrical Engineering students, Iran University of Science and Technology (2011).

Talks and Conference Presentations

Invited Talks and Seminars

- IMP–IMBA Young Investigator Symposium, Vienna, Austria (Apr. 2025).
- AI&A Journal Club, AstraZeneca, virtual (May 2022).

Oral Presentations

- Faculty Symposium, Department of Genetics, Washington University in St. Louis, St. Louis, MO (Jan. 2025).
- IGVF Consortium Meeting, Seattle, WA (Sept. 2024).
- Intelligent Systems for Molecular Biology (ISMB), Montreal, Canada (Jul. 2024).
- Biology of Genomes, Cold Spring Harbor Laboratory (May 2023).
- 4D Nucleome Scientific Webinar; virtual (Apr. 2022).

- Genome Informatics, Cold Spring Harbor Laboratory; virtual (Nov. 2021).
- ENCODE Consortium Meeting; virtual (Jan. 2021).
- ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB), Boston, MA (Aug. 2017).
- IEEE Global Conference on Signal and Information Processing (GlobalSIP), Orlando, FL (Dec. 2015).

Poster Presentations

- New York Symposium on Immune Cell Engineering and Reprogramming (Jul. 2026).
- AI in Biology Symposium, Cold Spring Harbor Laboratory (May 2026).
- Single Cell Genomics (SCG), Corinthia, Greece (Sept. 2024).
- Biology of Genomes, Cold Spring Harbor Laboratory (May 2024).
- 4D Nucleome Consortium Meeting, Boston, MA (Dec. 2023).
- Machine Learning in Computational Biology (MLCB), Seattle, WA (Nov. 2023).
- Memorial Sloan Kettering Postdoctoral Research Symposium, New York, NY; Best Poster Award (Sept. 2023).
- IGVF Consortium Meeting, St. Louis, MO; Best Poster Award (Sept. 2023).
- 4D Nucleome Consortium Meeting; virtual (Dec. 2021).
- Machine Learning in Computational Biology (MLCB); virtual (Nov. 2021).
- Biology of Genomes, Cold Spring Harbor Laboratory; virtual (May 2021).
- Machine Learning in Computational Biology (MLCB); virtual (Nov. 2020).
- IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), New Orleans, LA (Mar. 2017).

Professional Service

Journal peer review: Nature Communications; Genome Biology; Genome Research; IEEE Transactions on Signal Processing; IEEE/ACM Transactions on Computational Biology and Bioinformatics; BMC Bioinformatics; Journal of Applied Statistics.

Conference peer review: NeurIPS; ICML; ICLR; AAAI; IJCAI; RECOMB; Machine Learning in Computational Biology (MLCB).

Methods and Technical Expertise

Programming: Python and R; PyTorch and TensorFlow.

AI/ML: Transformers and foundation models; graph neural networks; neural ODEs and SDEs; variational inference and generative models; Bayesian learning and transfer learning.

Biological Data: Single-cell RNA-seq, ATAC-seq, and multiome; Perturb-seq; DNA sequence and epigenomic assays; Hi-C, HiChIP, and Micro-C.

References

Christina S. Leslie

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